

Forensic AI Investigation System

Setup & Installation Guide

Witness Audio → Transcript → Facial Attributes → AI Face Generation → Criminal Database Matching

Version	2.0 (Pure ONNX)
Python	3.10, 3.11, 3.12, 3.13, 3.14
Platform	Windows 10/11 (Linux/macOS compatible)
Install Size	~50 MB (API mode) / ~2.5 GB (Local mode)

Table of Contents

- 1. Requirements
- 2. Quick Start (5 Minutes)
- 3. Generation Modes
- 4. Default Login & Roles
- 5. The 4-Step Forensic Pipeline
- 6. Project Structure
- 7. Troubleshooting
- 8. Appendix: Manual Setup

1. Requirements

Requirement	Details
Operating System	Windows 10 or 11 (Linux/macOS also work)
Python	3.10 or higher (3.12 recommended)
Disk Space	~50 MB for API mode, ~2.5 GB for Local mode
Internet	Required for first setup and speech-to-text
RAM	4 GB minimum, 8 GB recommended for local mode

Install Python

If Python is not installed, download from:

<https://www.python.org/downloads/>

IMPORTANT: During installation, CHECK the box "Add Python to PATH".

2. Quick Start (5 Minutes)

1

Extract the project

Unzip the project folder to any location.

Example: `C:\forensic-ai\`

2

Double-click run.bat

The script will:

- Detect your Python version automatically
- Create a virtual environment
- Create a .env file from template
- Stop and ask you to fill in API keys

3

Edit the .env file

Open .env in Notepad and fill in your API keys (see table below).
Save the file.

4

Double-click run.bat again

The script will install all dependencies and start the server.

You will see: `Running on http://127.0.0.1:5000`

5

Open your browser

Go to `http://127.0.0.1:5000`

Login with: **superadmin / Admin@123**

Change this password immediately after first login.

API Keys for .env file:

Key	Required?	Where to get it
SECRET_KEY	Yes	Any long random string (30+ characters)
DEEPGRAM_API_KEY	Yes (for audio)	Free at https://console.deepgram.com/
HF_API_TOKEN	Only for API mode	https://huggingface.co/settings/tokens
GENERATION_MODE	Yes	"api" (lightweight) or "local" (offline)

3. Generation Modes

API Mode (Recommended for quick setup)

Set **GENERATION_MODE=api** in .env

- Uses FLUX.1-schnell via HuggingFace cloud API
- Lightweight install: only ~50 MB of packages
- Fast: ~10 seconds per face image
- Cost: ~\$0.003 per image (very cheap)
- Requires: HF_API_TOKEN + linked paid provider (fal-ai recommended)
- No sketch upload support

To link a paid provider: go to <https://huggingface.co/settings/inference-providers> and link fal-ai.

Local Mode (Recommended for offline/air-gapped)

Set **GENERATION_MODE=local** in .env

- Runs Stable Diffusion 1.5 on your machine
- First run downloads ~2.5 GB of model weights (one-time)
- Fully offline after first download
- Speed: ~3 minutes per image on CPU, ~15 seconds on GPU
- Supports witness sketch uploads (ControlNet)
- No API keys needed for generation (Deepgram still needs key for audio)

	API Mode	Local Mode
Install size	~50 MB	~2.5 GB
Speed per image	~10 seconds	~3 min (CPU)
Internet needed	Always	First run only
Cost	~\$0.003/image	Free
Sketch support	No	Yes
Best for	Quick setup	Offline use

4. Default Login & Roles

First Login

Username	superadmin
Password	Admin@123

Change this password immediately after first login.

User Roles

Role	Capabilities
Superadmin	Create/manage Admin accounts, view audit logs
Admin	Create Police officers, manage criminal database, rebuild face index
Police	Run the 4-step forensic pipeline, create investigation cases

Role Hierarchy

Superadmin → creates Admins → Admins create Police officers → Police run investigations

5. The 4-Step Forensic Pipeline

Login as a **Police** user and click **New Case** to start.

1

Upload Witness Audio

Record or upload a witness audio file (WAV, MP3, M4A, OGG, WEBM).
Deepgram Nova-2 transcribes it to text automatically.
Supports multiple languages with auto-detection.

2

Review Parsed Description

The system extracts facial attributes from the transcript:

- Gender, age, ethnicity, skin tone
- Hair color/style, eye color, facial hair
- Face shape, nose type, build
- Scars, moles, birthmarks (with precise location)
- Accessories: bindi, earrings, nose ring

You can edit the text and re-parse if needed.

3

Generate Candidate Faces

The AI generates 4 face images based on the description.

Stage 1: AI model creates the base face with accessories.

Stage 2: Scars and moles are rendered at precise locations using face landmark detection.

You can optionally upload a witness sketch for guided generation (local mode only).

4

Match Against Criminal Database

Select the best matching generated face.

The system compares it against the criminal database using:

- **ArcFace** structural similarity (face shape, proportions)
- **Attribute matching** (scars, gender, age, skin tone)

Top 5 matches are shown with confidence scores.

Before matching works, an Admin must add criminal records with face photos via the Admin panel and click 'Rebuild Face Database'.

6. Project Structure

File / Folder	Purpose
run.bat	Double-click to start (auto-detects Python, installs deps)
setup_check.py	Diagnostic script: python setup_check.py
app.py	Flask web application (all routes)
config.py	Configuration loader (reads .env)
.env	Your API keys (never share this file!)
.env.example	Template for .env
requirements.txt	Core Python packages (~50 MB)
requirements-local.txt	Additional packages for local mode (~2.5 GB)
modules/speech_to_text.py	Deepgram audio transcription
modules/attribute_parser.py	Witness text to structured attributes
modules/face_generator.py	Two-stage AI face generation
modules/face_enhancer.py	Scar/mole rendering via landmarks
modules/face_matcher.py	ArcFace two-layer matching
modules/onnx_face.py	Face detection + recognition (pure ONNX)
data/criminal_db/	Criminal face photos (added via Admin panel)
data/generated_faces/	AI-generated faces (overwritten each run)
data/forensic.db	SQLite database (auto-created)
database/schema.sql	Database schema
templates/	HTML templates for web UI

7. Troubleshooting

Symptom	Solution
run.bat closes instantly	Open Command Prompt, cd into the project folder, type run.bat to see the error
"Python was not found"	Install Python 3.10+ from python.org. CHECK 'Add Python to PATH'
First generation is very slow	Normal for local mode on CPU (~3 min/image). Use API mode for faster results
HuggingFace 402 error	Link a paid provider at huggingface.co/settings/inference-providers OR switch to GENERATION_MODE_API
No matches found	Criminal database is empty. Add criminals with photos via Admin panel, then click 'Rebuild Face Database'
Port 5000 in use	Close the other application using port 5000, or edit app.py to change the port
Module import errors	Run: <code>python setup_check.py --install</code>
Database corrupted	Delete data/forensic.db and restart. It recreates automatically

Diagnostic Command

Run this to check your entire setup:

```
python setup_check.py
```

To auto-install missing packages:

```
python setup_check.py --install
```

8. Appendix: Manual Setup

If run.bat does not work, follow these steps manually in Command Prompt:

Step 1: Open Command Prompt

```
Win+R → type cmd → press Enter
```

Step 2: Navigate to project folder

```
cd "C:\path\to\forensic-face-generation"
```

Step 3: Check Python version

```
python --version
```

Should show Python 3.10 or higher.

Step 4: Create virtual environment

```
python -m venv venv
```

Step 5: Activate virtual environment

```
venv\Scripts\activate
```

Step 6: Install core packages

```
pip install -r requirements.txt
```

Step 7: Install local mode packages (optional)

```
pip install -r requirements-local.txt
```

Only needed if GENERATION_MODE=local in .env. Skippable for API mode.

Step 8: Setup .env

```
copy .env.example .env
```

Edit .env in Notepad and fill in your API keys.

Step 9: Start the application

```
python app.py
```

Open browser: <http://127.0.0.1:5000>